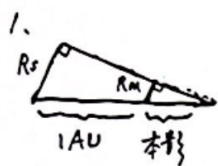




南京大学

NANJING UNIVERSITY

普通天文学作业5



$$\frac{x}{R_m} = \frac{x+1 \text{ AU}}{R_s} \Rightarrow x = \frac{R_m}{R_s - R_m} \text{ AU}$$

其中 $R_m = 1737.10 \text{ km}$, $R_s = 695500 \text{ km}$, $1 \text{ AU} = 1.496 \times 10^8 \text{ km}$

$$x = 3.7 \times 10^5 \text{ km}$$

注意到: 月球近地点距地球 $3.63 \times 10^5 \text{ km}$, 远地点距地 $4.06 \times 10^5 \text{ km}$.

故 ~~地球~~ 即使满足日食的轨道条件, 地球可能不处于本影中, 日全食的确罕见

2. 不妨设共线比为 $a:b = \frac{a}{b}$, $m:n = \frac{m}{n}$.

由 $\frac{a^3}{T^2} = \text{const}$, 得 $\frac{a_J^3}{T_J^2} = \frac{a^3}{(\frac{m}{M})^2 T_J^2}$, $a = (\frac{m}{M})^{\frac{2}{3}} a_J$

其中 $a_J = 7.79 \times 10^8 \text{ km} = 5.2 \text{ AU}$

3:1. $a = 2.5 \text{ AU}$ 3:2. $a = 3.27 \text{ AU}$

5:2. $a = 2.82 \text{ AU}$ 5:3. $a = 3.97 \text{ AU}$

7:3. $a = 2.96 \text{ AU}$

与 Kirkwood Gaps 在小数点后两位上十分吻合.

3. ~~空间站~~ 若宇航员脱离空间站 10 m. 取宇航员质量 100 kg.

$$F_H = G \frac{MM}{r^2} = [2.6 \times 10^{-7} \text{ m(kg)}] N = 2.6 \times 10^{-5} N \ll 1$$

所提供的速度 $2.6 \times 10^{-7} \text{ m/s}^2 \ll 1$.

即可忽略万有引力作用

$$\text{地球引力 } F_G = G \frac{MM_E}{R^2} = 2.5 \times 10^5 N$$

4.

带状区域体积 $V_1 = 150 \times 10^4 \times \pi (3 \text{ AU})^2 - (2 \text{ AU})^2 = 5.3 \times 10^{23} \text{ km}^3$

小行星总体积 $V_2 = 150 \times 10^4 \times \frac{4}{3} \pi (\frac{100}{2})^3 = 7.9 \times 10^{11} \text{ km}^3$

$$V_2/V_1 = 1/10^{12}$$

每颗小行星平均可自由运动的空间 $V_3 = \frac{V_2 V_1}{150 \times 10^4} = 3.5 \times 10^{17} \text{ km}^3$

忽略小行星半径, 平均距离 R 为 $(\frac{3.5 \times 10^{17} \times 3}{4\pi})^{1/3} = 4.4 \times 10^5 \text{ km}$.

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5. 系统初始总角动量 $L = \frac{mM}{m+M} d^2 \frac{2\pi}{T}$ $d = 1180 \text{ m}$ $T = 11^h 55^{\text{min}}$

碰撞改变系统角动量 $\Delta L = \frac{M}{M+m} d m_0 v_0$

则 $\frac{mM}{m+M} d^2 \frac{2\pi}{T} - \frac{M m_0}{M+m} d v_0 = \frac{(m+m_0)M}{m+m_0+M} d'^2 \frac{2\pi}{T} \Rightarrow d'^2 \frac{2\pi}{T} \frac{(m+m_0)(m+M)}{m+m_0+M} = m d \frac{2\pi}{T} - m_0 d v_0$

彗星初始总角动量方程: $\frac{GMm}{d^2} = m \left(\frac{2\pi}{T} \right)^2 \frac{Md}{m+M} \Rightarrow m+M = G \left(\frac{2\pi}{T} \right)^2 d^3$

碰撞后的角动量方程: $\frac{G(m+m_0)M}{d'^2} = (m+m_0) \left(\frac{2\pi}{T} \right)^2 \frac{Md'}{m+m_0+M}$

$d' = \left[G(m+m_0+M) \left(\frac{2\pi}{T} \right)^2 \right]^{1/3}$

$G(m+m_0+M)^{2/3} \left(\frac{2\pi}{T} \right)^{-4/3} \frac{2\pi}{T} \frac{(m+m_0)(m+M)}{m+m_0+M} = m d \frac{2\pi}{T} - m_0 d v_0$

$\left(\frac{2\pi}{T} \right)^{-1/3} = \frac{(m+m_0+M)^{1/3}}{(m+m_0)(m+M)} G^{-1/3} \left(m \frac{2\pi}{T} d - m_0 v_0 \right) d$

对于 M.M. 有 $m+M = G \left(\frac{2\pi}{T} \right)^2 d^3$

假定双小行星物质组成相同. 则 $\frac{M}{m} = \left(\frac{r}{R} \right)^3$

$\Rightarrow m = G \left(\frac{2\pi}{T} \right)^2 d^3 \frac{r^3}{r^3+R^3}$ $M = G \left(\frac{2\pi}{T} \right)^2 d^3 \frac{R^3}{r^3+R^3}$

代入解得 $T = 28800 \text{ s}$

$m = 4.48 \times 10^9 \text{ kg}$ $M = 5.20 \times 10^{10} \text{ kg}$

$\left(\frac{2\pi}{T} \right)^{-1/3} = 18.9$ $T' = 42419 \text{ s}$

$\Delta T = T' - T = 680 \text{ s}$

6. 太阳距地球质心的距离 $r = \frac{m_E}{m_S + m_E} 1 \text{ AU} = 4.5 \times 10^6 \text{ m}$

太阳绕质心运动线速度 $v = \frac{2\pi}{T} r = 0.895 \text{ m/s} \approx 0.9 \text{ m/s}$

太阳最大视向速度 $v_{\text{max}} = \frac{0.9}{\sqrt{2}} \text{ m/s} = 0.64 \text{ m/s}$

视向速度精度: 0.64 m/s

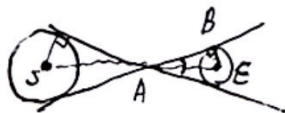
$\Delta \lambda = \lambda_0 \frac{2v_{\text{max}}}{c} = 656.3 \times \frac{0.64 \times 2}{3 \times 10^8} = 2.8 \times 10^{-6} \text{ nm}$

天体测量精度: $2r = 9 \times 10^6 \text{ m}$

$\Delta \theta = \frac{9 \times 10^6}{10^8} = 9 \times 10^{-2} \approx 6.03 \times 10^{-11} \text{ rad}$

7. 计算行星“半影的时间”

$\frac{AE}{AS} = \frac{R_E}{R_S}$ $AE + AS = \frac{a}{R_S} \Rightarrow AE = \frac{R_E}{R_S} a$
 $\theta = 2 \arcsin \frac{R_E}{R_S} = 2 \arcsin \frac{R_E}{R_S + R_E} = \frac{2(R_E + R_S)}{a}$



木星 $R_E = 7.1492 \times 10^7 \text{ m}$ $a = 7.79 \times 10^{10} \text{ m}$

木星半影 = $\frac{R_E + R_S}{\pi a} = 3.13\%$ 地球半影 = 1.48%

